PRAKTIKUM SISTEM OPERASI

MODUL 10

SHELL

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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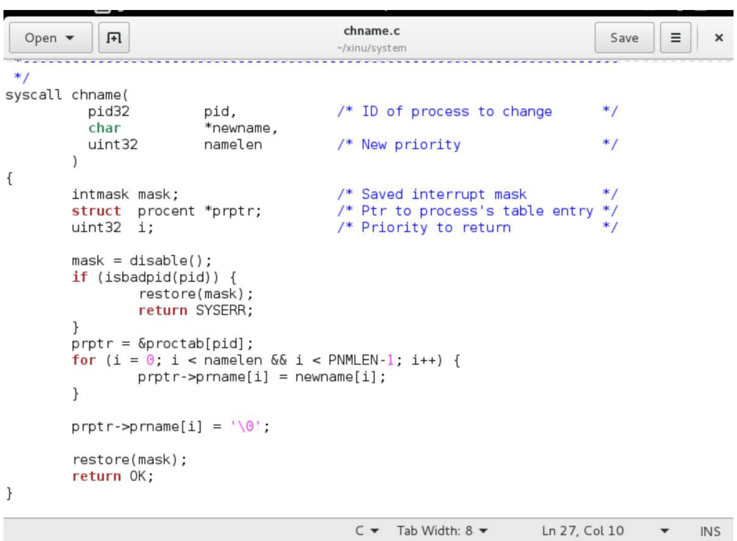
JURNAL/TP

1. Soal 1

Modifikasi syscall chname

Source code chname.c



Fungsi chname() digunakan untuk mengubah nama proses berdasarkan PID tertentu. Nama baru disalin ke field prname pada tabel proses menggunakan perulangan hingga seluruh karakter nama tersalin.

2. **Soal 2**

Pembuatan command baru namecmd

xsh_namecmd.c
~/xinu/shell

```c
/* NAMA : Zhafir Zaidan Avail */
/* NIM  : 23111040598*/

#include <xinu.h>

shellcmd xsh_namecmd(int nargs, char *args[])
{
	printf("my new command\n");

	/* pid 2 = netin */
	chname(2, "hellowww", 7);

	return 0;
}
```

Penambahan pada shprototypes.h

shprototypes.h
~/xinu/include

```c
/* in file xsh_ls.c */
extern  shellcmd  xsh_ls        (int32, char *[]);

/* in file xsh_led.c */
extern  shellcmd  xsh_led       (int32, char *[]);

/* in file xsh_memdump.c */
extern  shellcmd  xsh_memdump   (int32, char *[]);

/* in file xsh_memstat.c */
extern  shellcmd  xsh_memstat   (int32, char *[]);

/* in file xsh_namecmd.c */
extern  shellcmd  xsh_namecmd   (int32, char *[]);

/* in file xsh_netinfo.c */
extern  shellcmd  xsh_netinfo   (int32, char *[]);

/* in file xsh_ns.c */
extern  shellcmd  xsh_ns        (int32, char *[]);

/* in file xsh_nvram.c */
extern  shellcmd  xsh_nvram     (int32, char *[]);

/* in file xsh_ping.c */
extern  shellcmd  xsh_ping      (int32, char *[]);

/* in file xsh_ps.c */
```

C/C++/ObjC Header ▾ Tab Width: 8 ▾ Ln 49, Col 1 ▾ INS

Penambahan pada shell.c

```
shell.c
~/xinu/shell

	{"kill",	xsh_kill},
	{"memdump",	xsh_memdump},
	{"memstat",	xsh_memstat},
	{"ns",		xsh_ns},
	{"netinfo",	xsh_netinfo},
	{"ping",	xsh_ping},
	{"ps",		xsh_ps},
	{"sleep",	xsh_sleep},
	{"tee",		xsh_tee},
	{"udp",		xsh_udpdump},
	{"udpecho",	xsh_udpecho},
	{"udpeserver",	xsh_udpeserver},
	{"uptime",	xsh_uptime},
	{"namecmd",	xsh_namecmd},
	{"?",		xsh_help}

};

uint32	ncmd = sizeof(cmdtab) / sizeof(struct cmdent);


/************************************************************************/
/*									*/
/* cmdlookup -- return the index in cmdtab of a command name		*/
/*									*/
/************************************************************************/

int32	cmdlookup(char *name) {
```

3. Pengujian Program

```
xinu@xinu-develop-end: ~/xinu/compile
---------------2311104059------------------


Welcome to Xinu!


xsh $ ps
Pid Name             State Prio Ppid Stack Base  Stack Ptr  Stack Size Msg Content
--- ---------------- ----- ---- ---- ---------- ---------- ---------- --- -------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFEE4       8192  1       3
  1 ipout            wait   500    0 0x005FBFFC 0x005FBF30       8192  0       0
  2 netin            wait   500    0 0x005F9FFC 0x005F9ED0       8192  0       0
  4 Main process     recv    20    3 0x005E7FFC 0x005E7F34      65536  0       0
  5 shell            recv    50    4 0x005F7FFC 0x005F79AC       8192  0       0
  6 ps               curr    20    5 0x005F5FFC 0x005F5FC4       8192  0       0
xsh $ namecmd
my new command
xsh $ ps
Pid Name             State Prio Ppid Stack Base  Stack Ptr  Stack Size Msg Content
--- ---------------- ----- ---- ---- ---------- ---------- ---------- --- -------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFEE4       8192  1       3
  1 ipout            wait   500    0 0x005FBFFC 0x005FBF30       8192  0       0
  2 helloww          wait   500    0 0x005F9FFC 0x005F9ED0       8192  0       0
  4 Main process     recv    20    3 0x005E7FFC 0x005E7F34      65536  0       0
  5 shell            recv    50    4 0x005F7FFC 0x005F79AC       8192  0       7
  8 ps               curr    20    5 0x005F5FFC 0x005F5FC4       8192  0       0
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:7 | ttyS0
```

Berdasarkan praktikum yang telah dilakukan, syscall chname berhasil dimodifikasi sehingga dapat digunakan untuk mengubah nama suatu proses berdasarkan PID yang diberikan. Selain itu, command baru bernama namecmd berhasil ditambahkan ke shell Xinu dan dapat digunakan untuk memanggil syscall chname. Hasil pengujian menunjukkan bahwa nama proses berhasil berubah dari netin menjadi helloww, sehingga tujuan praktikum Modul 10 dapat tercapai dengan baik.